Divide-and-Conquer Sorting and Recurrences

- Merge Sort: Divide-and-conquer sort that merges sorted halves in $O(n \log n)$ worst-case time.

- Recurrence for Merge Sort: $T(n) = 2T(n/2) + O(n)$

- Heapsort: In-place comparison sort using a binary max-heap in $O(n \log n)$ time and $O(1)$ space.

- Quicksort: Partition-based sort running in $O(n \log n)$ expected time under random pivots.

- Partition balance increases when randomized pivot selection decreases worst-case $O(n^{2})$ splits.

Binary Search, Asymptotic Bounds, and Pseudocode

- Binary Search: Halves a sorted search interval each iteration in $O(\log n)$ worst-case time.

- Big-O Bound: Function $f(n)$ is $O(g(n))$ when $|f(n)| \le c|g(n)|$ for all $n \ge n_{0}$.

- Big-Theta Bound: Tight asymptotic bound where $f(n)$ is bounded above and below by $g(n)$.

- Loop Invariant: Subarray A[low..high] contains target whenever target exists in sorted array A.

- Binary Search Step: Compute midpoint m = (low + high) // 2 and compare A[m] against key.

Graph Algorithms and Shortest-Path Complexity

- Breadth-First Search: Explores vertices level by level from a source in $O(V + E)$ time via a queue.

- Depth-First Search: Recursively explores incident edges to classify tree, back, and cross edges.

- Dijkstra Algorithm: Computes nonnegative single-source shortest paths in $O((V + E) \log V)$ time.

- Topological Sort: Linear ordering of vertices in a directed acyclic graph respecting all edge directions.

Calculus Rules for Continuous Optimization

- Quotient Rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^{2}}$

- Chain Rule: Differentiates composite functions via $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$.

- Fundamental Theorem: Evaluates definite integrals via $\int_{a}^{b} f(x)\,dx = F(b) - F(a)$.

- Gradient Descent: Updates parameters via $x_{k+1} = x_{k} - \alpha \nabla f(x_{k})$ along steepest descent.